

Understanding Cholesterol

Diet, Lipid Panel, and Lifestyle — Filipino Context

Companion to the online guide

LIPID PANEL QUICK REFERENCE

Reading Your Lipid Panel

Measurement	What It Reflects	Desirable	Borderline	High / Low Risk
Total Cholesterol (TC)	Sum of all cholesterol particles. Screening value — interpret with LDL & HDL.	<200 mg/dL	200–239	≥240
LDL-Cholesterol (LDL-C)	Primary atherogenic driver — main target of lipid-lowering therapy.	<100 mg/dL	100–129	≥130 (high risk: <70)
HDL-Cholesterol (HDL-C)	Protective particle — higher is better. Low HDL is an independent risk factor.	≥60 mg/dL	40–59 M / 50–59 F	<40 M / <50 F
Triglycerides (TG)	Reflects carb & fat intake, alcohol use, and insulin resistance.	<150 mg/dL	150–199	≥200 (≥500 = pancreatitis risk)
Non-HDL-C (TC – HDL)	Captures all atherogenic particles (LDL + VLDL + IDL). Often more predictive than LDL alone.	<130 mg/dL	130–159	≥160
TG / HDL Ratio	Marker for small-dense LDL and insulin resistance. Calculate: TG ÷ HDL.	<3.5	3.5–5.0	>5.0 (IR likely)

⚠ Your LDL target depends on your cardiovascular risk category

If you have heart disease, diabetes, CKD, or multiple risk factors, your target LDL-C may be <70 mg/dL or even <55 mg/dL. Ask your physician for your individual target based on your full risk profile.

DIETARY FAT TYPES

How Different Fats Affect Your Lipids

Fat Type	LDL-C	HDL-C	TG	Common Filipino / Philippine Sources
Saturated Fat (SFA)	↑ Raises	Neutral	Neutral	Coconut oil (gata), lard (mantika), butter, fatty pork (liempo, lechon), chicharon, full-fat dairy
Trans Fat (TFA)	↑↑ Raises	↓ Lowers	Neutral	Partially hydrogenated margarine, commercial pastries, some imported biscuits & cookies — avoid entirely
Monounsaturated (MUFA)	↓ Lowers	Neutral	Neutral	Olive oil, canola oil, avocado, pili nuts, cashews, almonds
Omega-6 PUFA	↓ Lowers	Neutral	Neutral	Sunflower oil, corn oil, vegetable oil blends
Omega-3 PUFA (EPA/DHA)	Neutral	↑ Raises	↓↓ Lowers	Sardinas, galunggong, bangus, salmon, canned tuna — 2 servings/week
Soluble Fiber	↓ Lowers	Neutral	Neutral	Oatmeal, monggo, okra, kamote, psyllium husk — traps bile acids; 5–10 g/day reduces LDL ~5–11 mg/dL

Coconut oil is 92% saturated fat

Despite popular claims, coconut oil raises LDL-C consistently in clinical trials. Major cardiology guidelines (AHA, ACC) recommend replacing coconut oil and lard with **canola oil or olive oil** as your primary cooking fat. For occasional use in traditional dishes, small amounts are acceptable — but not as an everyday staple.

FILIPINO FOOD GUIDE

Foods to Limit or Avoid — Philippine Context

AVOID Very High Saturated / Trans Fat

- **Chicharon** — ~40 g contains 12–15 g saturated fat
- **Commercial fried fast food** — battered chicken, fries in palm oil
- **Processed meats** — low-quality hotdog, Vienna sausage with high fat fillers
- **Commercial pastries** — ensaymada, cream-filled biscuits
- **Partially hydrogenated margarine** — trans fat; replace with canola or olive oil

LIMIT Moderate Sat. Fat — Occasional OK

- **Lechon, liempo, crispy pata** — special occasions, small portions only
- **Kare-kare, laing, ginataan** — each serving = 15–25 g sat. fat; use light coconut milk
- **Longganisa & tocino** — high in sat. fat + sodium; prefer homemade lean versions
- **Full-fat dairy** — cheese, cream; replace with low-fat or skim
- **Egg yolks** — 1–2 per day acceptable; limit to 1/day if LDL is elevated
- **Large white rice portions** — drive TG via de novo lipogenesis; moderate portion size

FILIPINO FOOD GUIDE

Heart-Healthy Foods to Embrace — Available at Any Wet Market

EAT Oily Fish — Omega-3 Rich

- **Sardinas** (canned, in water/tomato sauce) — affordable EPA/DHA
- **Galunggong** — best value omega-3 source in PH markets
- **Bangus** — grilled or steamed; avoid deep-fried
- **Tuyo** — good omega-3 but high sodium; once weekly max
- Target: **2 servings of oily fish per week**

EAT Legumes — Soluble Fiber & Protein

- **Monggo (mung beans)** — guisado form is a heart-healthy staple
- **Kadyos (pigeon peas)** — replaces fatty meat in soups
- **Tokwa (firm tofu)** — plant protein; replace pork in dishes
- **Black beans, red kidney beans** — add to soups and stews

EAT Vegetables

- **Kangkong** — inexpensive, high fiber; sauté with minimal oil
- **Okra** — mucilage traps bile acids; excellent LDL-lowering
- **Ampalaya** — modest LDL reduction + blood sugar benefit
- **Malunggay** — fiber and antioxidants; add to soups freely
- **Kamote** — soluble fiber, lower glycemic index than rice

EAT Grains, Nuts & Oils

- **Rolled oats / oatmeal** — β -glucan lowers LDL ~5–8% at 70–100 g dry/day
- **Brown rice** — more fiber than white; replace at least half your rice intake
- **Pili nuts** — native to Bicol; high MUFA/PUFA; 30 g/day is cardioprotective
- **Cashews** — local, accessible; mostly MUFA; 1 small handful/day
- **Canola oil** — best all-purpose cooking oil; high MUFA, low SFA
- **Extra-virgin olive oil** — for dressings and low-heat cooking
- **Garlic** — lowers LDL ~3–5%; use liberally

✓ **Sample Heart-Healthy Filipino Day**

Breakfast	Oatmeal + sliced banana + 1 boiled egg (or tokwa scramble) + black coffee / unsweetened tea
Lunch	Brown rice + monggo guisado with malunggay + grilled galunggong + tomato ensalada
Dinner	Sardines in tomato sauce + steamed kangkong with garlic + small brown rice portion
Snack	Handful pili nuts or cashews + fresh fruit (not juice)

LIFESTYLE & NON-DIETARY FACTORS

Lifestyle Changes That Move the Lipid Panel

Aerobic Exercise

↑ **HDL 3–9%** · ↓ **TG 10–20%**

150 min/week of brisk walking, swimming, or cycling. Even shifts LDL from small-dense (harmful) to larger, less atherogenic particles.

Weight Loss

↓ **TG 20–30%** · ↑ **HDL 5–8%**

Losing 5–10% of body weight measurably improves all lipid values. Every kilogram lost provides benefit — ideal weight not required.

Quit Smoking

↑ **HDL within weeks**

Smoking lowers HDL by 4–6 mg/dL and promotes LDL oxidation. No medication can offset the damage from active smoking.

Limit Alcohol

↓ **TG 30–50% if stopped**

Any alcohol raises triglycerides. If TG is elevated, reducing or stopping alcohol is often the fastest single dietary intervention.

Reduce Refined Carbs

↓ **TG fast** · ↓ **TG/HDL ratio**

Replace white rice, white bread, sugary drinks, and sweet snacks. Carb reduction often lowers TG faster than fat restriction in insulin-resistant patients.

Sleep & Stress

Modest lipid benefit

Poor sleep (<6 h/night) raises LDL and TG independently. Chronic stress elevates cortisol, increasing hepatic fat synthesis.

Educational use only. This handout does not replace individualized medical advice. Lipid targets and medication decisions depend on your full cardiovascular risk profile — consult your physician before making dietary or medication changes. References: ACC/AHA 2025 Lipid Guidelines · AHA 2021 Dietary Fats Advisory · KDIGO 2024 Lipids in CKD.

[williamriveromd.com](https://williamriveromd.com/guides/cholesterol-diet-guide.html)
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